Ex.No : 02 RECOVERY AND ANALYSIS OF DELETED FILES FROM A USB DRIVE

Date :

AIM:

To recover deleted files from a USB drive using a digital forensic tool and analyze the recovered data for forensic investigation.

SOFTWARE/EQUIPMENTS REQUIRED:

- Windows 10/11 or Linux (Ubuntu/Kali)
- TestDisk , PhotoRec

THEORY:

Deleted files are not immediately removed from a storage device. When a file is deleted, the operating system typically marks its storage space as available for reuse, while the actual data remains on the device until it is overwritten. Digital forensic tools can recover these files by scanning the file system or performing signature-based recovery. File recovery is an essential process in digital forensics, cybercrime investigations, and incident response.

TOOL USED:

PhotoRec is an open-source command-line data recovery tool used to recover deleted files from storage devices such as USB drives, hard disks, memory cards, and SSDs. It uses file signature-based recovery and supports a wide range of file systems, including FAT, NTFS, exFAT, ext2/ext3/ext4, HFS+, and others. It is widely used in digital forensics to recover deleted or lost files from damaged or formatted storage media.

ALGORITHM:

1. Start the experiment and ensure that TestDisk/PhotoRec is installed on the system.
2. Connect the test USB drive and identify its device name and storage capacity.
3. Launch PhotoRec with the required administrative privileges.
4. Select the USB drive from the list of available storage devices.
5. Select the appropriate partition and filesystem type.
6. Choose the Free option to scan unallocated space for deleted files, or Whole to scan the entire partition.
7. Select a destination directory on a different storage device to store the recovered files.
8. Start the recovery process and wait for PhotoRec to complete the scan.
9. Examine the recovered files and record the file types, recovery count, and recovery status.
10. Document the observations and forensic findings.
11. End

PROCEDURE

Step 1: Download PhotoRec

1. Download the Windows version of TestDisk.
2. Extract the downloaded ZIP file to a folder such as C:\DMF\Exp 2\TestDisk.
3. Open the extracted folder.
4. Locate photorec_win.exe.

Step 2: Prepare a test USB drive

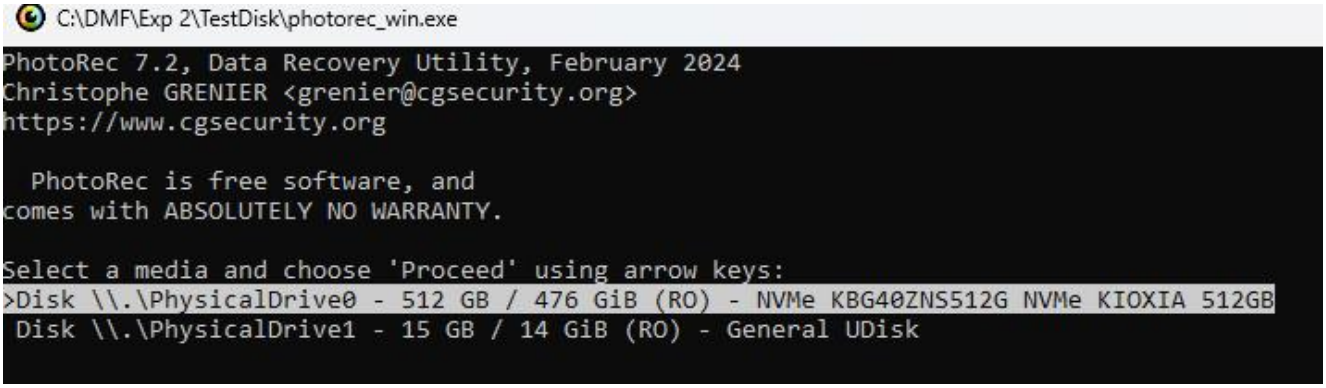
1. If you do not already have deleted sample files, create a practice dataset first.
2. Connect a USB drive to your computer.
3. Create a folder named DMF_Test.
4. Copy a few sample files into it, such as: sample.txt, sample.jpg, sample.pdf
5. Open each file once to confirm it works.
6. Delete these test files and empty the Recycle Bin if prompted.
7. Do not copy new files onto the USB drive after deleting them. Recovery may fail if the deleted data has been overwritten.

Step 3: Launch PhotoRec

1. Open the extracted TestDisk folder.
2. Right-click photorec_win.exe.
3. Select Run as administrator.
4. Approve the Windows permission prompt.

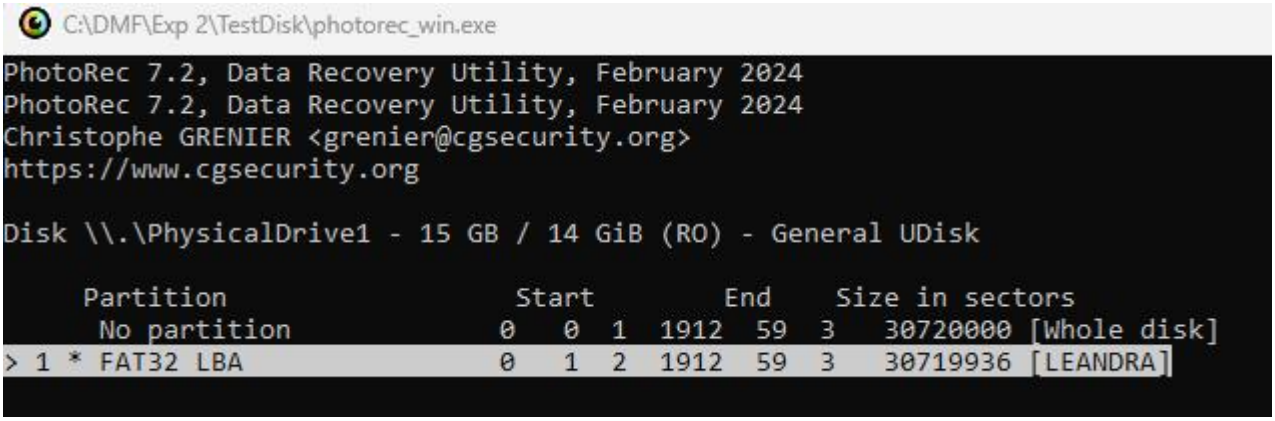
Step 4: Select the USB drive

1. Use the Up/Down arrow keys to select your USB drive.
2. Identify it by its storage capacity. For example, a 16 GB USB drive should be distinguishable from your computer's internal drive.
3. Press Enter to proceed.



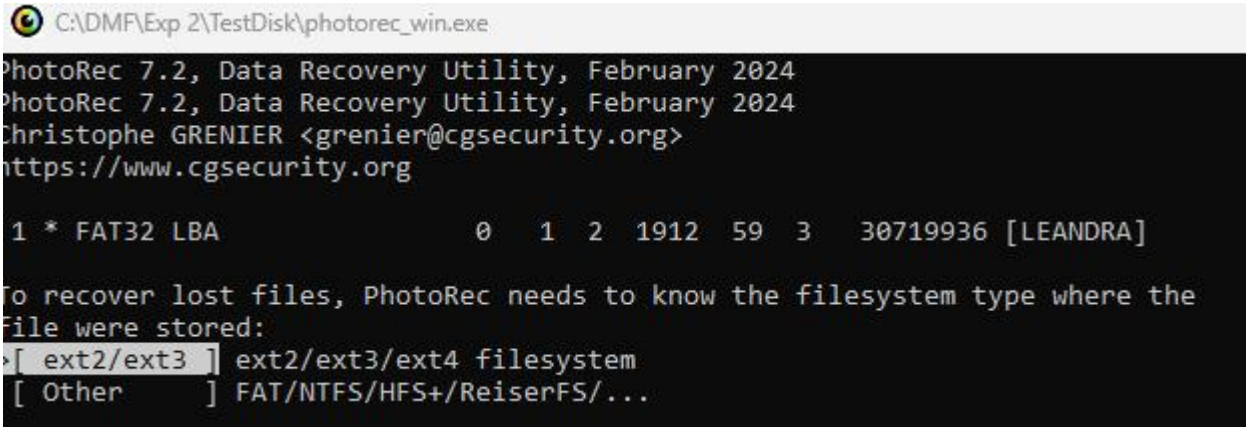
Step 5: Select the partition

1. Select the partition belonging to your USB drive.
2. Choose Search and press Enter.
3. If the partition is not shown as expected, stop and check that you selected the correct device.



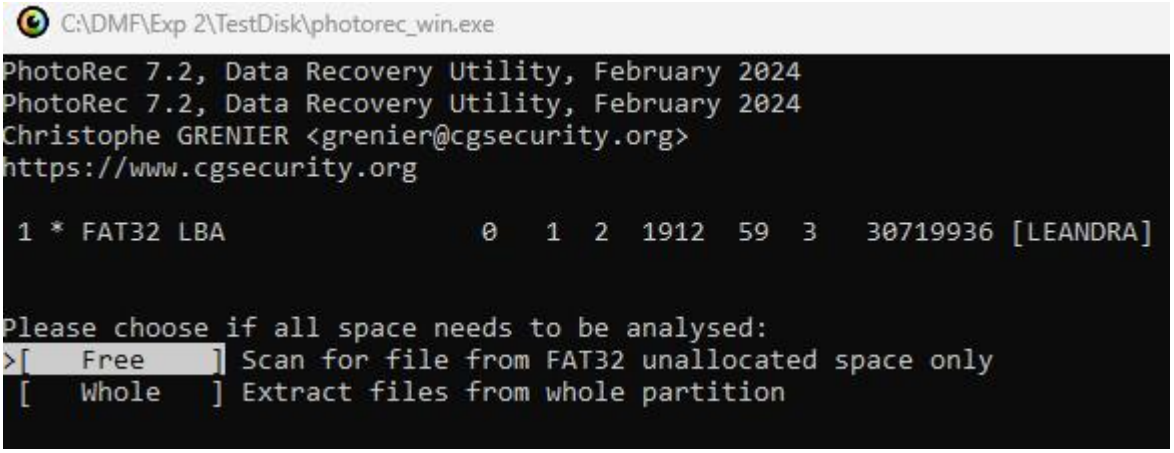
Step 6: Select the filesystem type

1. PhotoRec will ask which filesystem family was used.
2. Select Other for common Windows USB filesystems such as FAT32, exFAT, and NTFS.
3. Select the ext2/ext3/ext4 option only if the partition uses a Linux filesystem.



Step 7: Choose the scan area

1. PhotoRec will offer scanning choices.
2. Free — scans unallocated space for deleted files, where supported.
3. Whole — scans the entire selected partition.

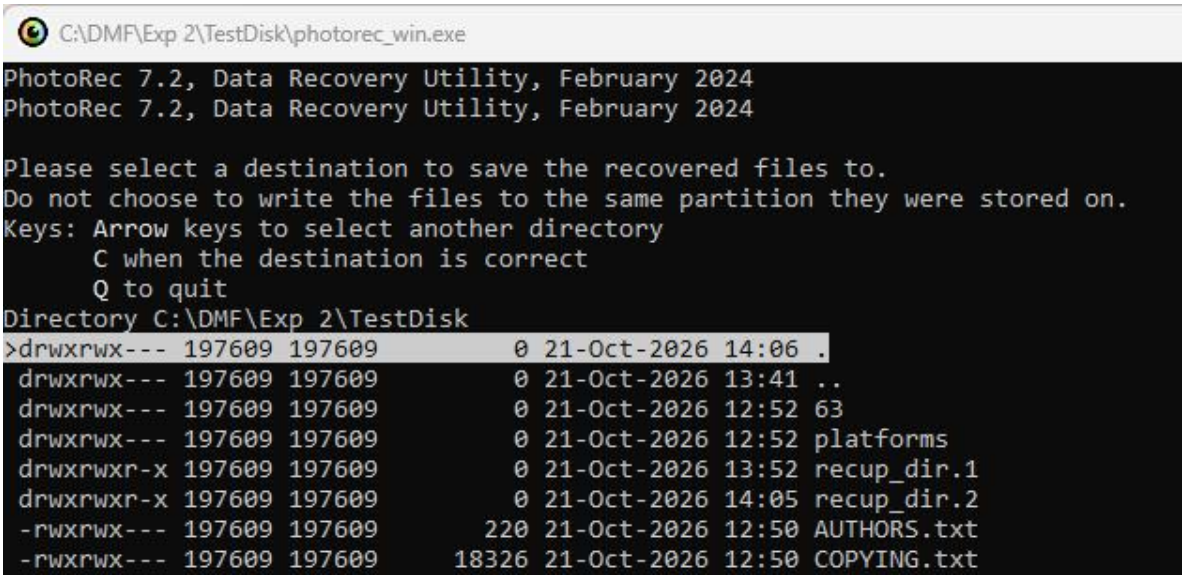


Step 8: Choose the recovery destination

1. Create a folder on your computer, for example C:\DMF\Recovered.
2. In PhotoRec, use the arrow keys to navigate to the destination folder.
3. Press Enter to enter folders.

4. When the correct destination is selected, press the key indicated by PhotoRec to confirm it (commonly C in the documented interface).

5. Never select the USB drive being scanned as the destination. Recovered files must be written to a different drive or filesystem.



Step 9: Recover the files

1. PhotoRec starts scanning the selected area, Wait for the scan to complete.
2. Observe the recovery count displayed on the screen.
3. After completion, exit PhotoRec.
4. The recovered files are normally saved in folders named recup_dir.1, recup_dir.2, and so on, inside the destination folder. The original filenames and folder structure may not be preserved.

Output:

